

# Does the Row IV $1/\bar{\epsilon}$ feed the running coupling?

Four tests, and where  $\beta_0$  actually lives

$$\beta_0 = \frac{11N_c}{3} - \frac{2n_f}{3}, \quad \mathcal{P}_{gg}(\xi) = 2N_c \left[ \frac{\xi}{[1-\xi]_+} + \frac{1-\xi}{\xi} + \xi(1-\xi) \right] + \frac{\beta_0}{2} \delta(1-\xi), \quad L_r = \log(4\pi^2 \mu^2 r^2)$$

**No — not this pole.** The shape is deceptively close, so it is worth being careful, but all four tests come out the same way. A running-coupling term at  $O(g^4)$  must be  $g^4 \beta_0 \log(\mu^2 r^2) \times (\text{LO})$ : a *pure*  $\delta(1-\xi)$ , with *no* rapidity logarithm, from a *ultraviolet* pole, carrying  $n_f$ . Row IV's  $\log(4\pi^2 \mu^2 r^2)$  instead arrives multiplied by  $\delta(1-\xi) \log \frac{k^+}{\Lambda} + \frac{\mathcal{P}_{gg}(\xi)}{2N_c}$ , from a pole at  $|z| \rightarrow \infty$ , in a row with no quark loop.  $\beta_0$  lives in the gluon self-energy / vertex rows and in the  $g \rightarrow q\bar{q}$  row, none of which is this one.

## 1. Order counting: what an $O(g^4)$ running-coupling term must look like

The LO cross section is  $O(g^2)$  and this row is  $O(g^4)$ , so the only way a coupling can run inside this calculation is by absorbing a  $\log \mu^2$  into the LO coupling:

$$g^2(\mu_0) = g^2(\mu) \left[ 1 + \frac{\alpha_s \beta_0}{4\pi} \log \frac{\mu^2}{\mu_0^2} + \dots \right] \implies \underbrace{g^2}_{\text{LO}} \rightarrow \underbrace{g^4 \beta_0 \log(\mu^2 r^2)}_{O(g^4)} \times (\text{LO structure}).$$

Read off the three properties that term must have. It is multiplied by the **LO structure and nothing else** — so a pure  $\delta(1-\xi)$ , no  $\xi$ -dependence, no  $\mathcal{P}_{gg}$ . It carries **no rapidity logarithm** — the LO cross section has no  $\log \frac{k^+}{\Lambda}$  to attach one to. And its coefficient is  $\beta_0$ .

Row IV's bracket is

$$\left[ \delta(1-\xi) \log \frac{k^+}{\Lambda} + \frac{\mathcal{P}_{gg}(\xi)}{2N_c} \right] \left[ \frac{1}{\bar{\epsilon}} - \log(4\pi^2 \mu^2 r^2) \right] \mathcal{X},$$

and neither half fits:

- The  $\delta(1-\xi)$  half carries an *extra*  $\log \frac{k^+}{\Lambda}$ . That makes it  $\bar{\alpha}_s \log \frac{k^+}{\Lambda} K \otimes (\text{LO})$  — the JIMWLK evolution term. Making the coupling *in that kernel* run is a correction to  $\bar{\alpha}_s K$ , i.e.  $\bar{\alpha}_s^2 \beta_0 \log \times K \otimes (\text{LO})$ , which is  $O(g^6)$  — one full order beyond this calculation. Running-coupling BK is a real and important effect, but it is not at this order.
- The  $\mathcal{P}_{gg}(\xi)$  half is  $\xi$ -dependent. A  $\beta$  function has no  $\xi$ ; a splitting kernel does. They cannot be the same object.

## 2. The $\beta_0$ coefficient test: what $\frac{11N_c}{6}$ hangs off

It is worth seeing exactly where  $\frac{11N_c}{6}$  comes from, because that is what tells you it is not here. Impose the momentum sum rule  $\int_0^1 dz z \mathcal{P}_{gg}(z) + 2n_f \int_0^1 dz z \mathcal{P}_{qg}(z) = 0$  with  $\mathcal{P}_{qg} = T_R[z^2 + (1-z)^2]$ :

| piece               | $\int_0^1 dz z \times$ |
|---------------------|------------------------|
| $\frac{z}{[1-z]_+}$ | $-\frac{3}{2}$         |
| $\frac{1-z}{z}$     | $+\frac{1}{2}$         |
| $z(1-z)$            | $+\frac{1}{12}$        |
| sum, $\times 2N_c$  | $-\frac{11N_c}{6}$     |

so the  $\delta(1-z)$  coefficient must be  $+\frac{11N_c}{6} - \frac{2T_R n_f}{3} = \frac{11N_c}{6} - \frac{n_f}{3} = \frac{\beta_0}{2}$  (sympy,  $T_R = \frac{1}{2}$ ). Notice which entry carries it: the whole  $\frac{11}{6}$  hangs off the  $-\frac{3}{2}$ , i.e. off the **soft endpoint**  $z \rightarrow 1$  of  $\frac{z}{[1-z]_+}$ . Whatever regulates that endpoint decides what sits on  $\delta(1-\xi)$ .

In Row IV the soft endpoint is cut off **in rapidity**: the  $p^+$  integral runs  $\Lambda < p^+ < \vee - k^+$ , and the plus prescription returns  $\delta(1-\xi) \log \frac{k^+}{\Lambda}$ . The constant  $\frac{11N_c}{6}$  would require that endpoint to be cut off by *transverse* kinematics instead. So this row does not, and cannot, put  $\beta_0$  on its  $\delta(1-\xi)$  — it puts a rapidity logarithm there.

### 3. The $n_f$ test

$\beta_0 = \frac{11N_c}{3} - \frac{2n_f}{3}$ . The  $n_f$  piece comes from the  $g \rightarrow q\bar{q}$  row and from nowhere else. Row IV has no quark loop, so at best it could supply part of the  $\frac{11N_c}{3}$  — and that partial sum is not a running coupling until the  $-\frac{2n_f}{3}$  shows up beside it in the same  $\log(\mu^2 r^2)$ .

**The one test that settles it, when you have more rows.** Collect the coefficient of  $\delta(1-\xi) \log(\mu^2 r^2) \mathcal{XU}$  — with *no*  $\log \frac{k^+}{\Lambda}$  attached — from every row, including the quark-loop one. If the gluonic rows sum to  $\frac{11N_c}{3}$  and the quark row supplies  $-\frac{2n_f}{3}$ , that combination is  $\beta_0$  and it is absorbed by  $\alpha_s(\mu) \rightarrow \alpha_s(1/r^2)$  in the LO cross section. Row IV's contribution to that particular coefficient is **zero**, because its  $\log(\mu^2 r^2)$  never appears without either  $\log \frac{k^+}{\Lambda}$  or  $\mathcal{P}_{gg}(\xi)$  beside it.

### 4. The region test: a coupling is renormalised by a UV pole

This is the cleanest of the four. Coupling renormalisation absorbs a *short*-distance divergence. Row IV's pole is at *long* distance. From the power counting, with  $u = y - z$ ,  $|u| = \rho$ :

$$\underbrace{\frac{u^m(u+r)^{m'}}{u^2(u+r)^2} \sim \frac{1}{\rho} \frac{1}{|r|}, \quad d^2z = \rho d\rho d\phi \implies \int_0^{\rho} \frac{d\rho}{|r|} \text{ finite}}_{z \rightarrow y, \text{ no UV pole}} \quad \underbrace{\sim \frac{1}{z^2}, \quad d^2z = |z|d|z|d\phi \implies \int^R \frac{d|z|}{|z|} = \log R}_{|z| \rightarrow \infty, \text{ the pole}}$$

The single power of  $u^m$  in the numerator is what kills the short-distance divergence. So there is no UV pole in  $\mathcal{G}^{mm'}$  to renormalise anything with.

### 5. Summary table

|                                 | Row IV's pole                                                           | a running-coupling pole                                 |
|---------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------|
| <b>region</b>                   | $ z  \rightarrow \infty$ (infrared)                                     | $z \rightarrow y$ (ultraviolet)                         |
| $n_f$                           | absent — no quark loop in this row                                      | must appear, as $-\frac{2n_f}{3}$                       |
| $\delta(1-\xi)$ coefficient     | $\log \frac{k^+}{\Lambda}$ (rapidity)                                   | $\frac{11N_c}{6}$ (a number, no log)                    |
| $\xi$ -dependence               | yes — $\mathcal{P}_{gg}(\xi)$ rides the same bracket                    | none — pure LO structure                                |
| <b>fate of <math>\mu</math></b> | appears only in $\frac{1}{\epsilon} - L_r$ , so it leaves with the pole | survives as $\alpha_s(\mu) \rightarrow \alpha_s(1/r^2)$ |
| <b>colour structure</b>         | LO, $\mathcal{XU}$ , in both cases — <i>not</i> a discriminator         |                                                         |

The last row is the reason the question is a good one: the colour and transverse structure of the two really do look alike, and  $\log(\mu^2 r^2) \times (\text{LO structure})$  is exactly the fingerprint of  $\alpha_s(1/r^2)$ . The five other rows are what separate them.

**Consistency with what I said last time.** If the  $\frac{1}{\epsilon}$  cancels against the companion Group II rows, then  $\log \mu^2$  cancels *with* it — they occur only in the single combination  $\frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2)$ . So there is no leftover  $\mu$ -dependence in this row for a running coupling to absorb, which is the same conclusion reached from the other direction. These two readings agree; they are not alternatives.

Checks: `src/rowG2_beta0.py` (the momentum sum rule giving  $\frac{11N_c}{6} - \frac{n_f}{3}$ , the order counting, the  $n_f$  and region tests), `src/rowG2_advisor.py` (the single-pole coefficient and the  $z \rightarrow y$  power counting), `src/rowG2_fate.py` (the BK identity).